# تقرير التحليل التجاري

Planted sample

| | |
|---|---|
| **الإصدار** | 1 |
| **فترة البيانات** | 2023-01-06 – 2024-12-31 |
| **الصفوف** | 21,971 |
| **درجة جودة البيانات** | 100 / 100 |
| **العملة** | غير محددة |
| **تشغيل التحليل** | 2026-11-05 |
| **تاريخ الإنشاء** | 2026-11-05 |

كل رقم في هذا التقرير مأخوذ من النتائج المحفوظة لتشغيل التحليل المذكور أعلاه: استعلامات SQL ونماذج إحصائية نُفذت على بياناتك. لم يُقدَّر أي رقم أو يُكتب بواسطة نموذج ذكاء اصطناعي. لا يتضمن التقرير سيناريوهات ماذا لو.

عناوين النتائج وملاحظاتها يكتبها محرك التحليل باللغة الإنجليزية.

## الملخص التنفيذي

النتائج ذات الأثر التجاري المقدَّر الأكبر، من الأعلى إلى الأدنى.

**1. Year to date is 196.7% ahead of last year**

Net revenue from 2024-01-01 to 2024-12-31 is 9,112,029.74, versus 3,071,272.26 for 2023-01-01 to 2023-12-31 (196.7%).

الثقة: عالية · العينة: 21,971 · درجة الأثر: 100

**2. Lasting downward shift in orders from 2024-03-11**

Weekly orders stepped down by about 46.8% from the week of 2024-03-11 and stayed at the new level for 23 weeks (compared with the same weeks a year earlier, so the normal season is taken out). That is roughly -2,574,048.61 of revenue since. These are findings from the data, not proven causes.

الثقة: عالية · العينة: 103 · درجة الأثر: 84

**3. Net revenue for Furniture is estimated at about 1,339,349.34 next month (80% range: 588,591.74 to 2,090,106.93)**

It is an estimate, not a promise: Net revenue for Furniture is estimated at about 1,339,349.34 for Jan 2025 (80% range: 588,591.74 to 2,090,106.93; 95% range: 191,164.68 to 2,487,534.00). Over the next 6 months the estimated total is about 8,603,653.42. Method: exponential smoothing (ETS), chosen by testing on past data (3 rolling windows of 5 months). In those tests its estimates were off by about 14.9% on average (sMAPE) and by about 89,093.02 per period (MAE); its error was 0.32 times that of repeating the previous cycle (MASE, below 1 is better). The seasonal naive baseline was off by about 96.4% (MASE 1.07). Treat it as a guide rather than a firm number.

الثقة: متوسطة · العينة: 7,002 · درجة الأثر: 60

**4. Net revenue is estimated at about 2,295,213.14 next month (80% range: 1,105,837.97 to 3,484,588.30)**
It is an estimate, not a promise: net revenue is estimated at about 2,295,213.14 for Jan 2025 (80% range: 1,105,837.97 to 3,484,588.30; 95% range: 476,220.79 to 4,114,205.48). Over the next 6 months the estimated total is about 14,674,100.06. Method: exponential smoothing (ETS), chosen by testing on past data (3 rolling windows of 5 months). In those tests its estimates were off by about 14.6% on average (sMAPE) and by about 153,425.60 per period (MAE); its error was 0.30 times that of repeating the previous cycle (MASE, below 1 is better). The seasonal naive baseline was off by about 98.4% (MASE 1.07). Treat it as a guide rather than a firm number.

الثقة: متوسطة · العينة: 21,971 · درجة الأثر: 60

**5. Champions customers bring 50.8% of revenue**
1,363 customers fall into 7 RFM segments. Champions bring the most revenue: 50.8% from 328 customers. 437 customers are At risk or Hibernating; they used to spend about 1,751,622.83 a year. Suggestion: Win-back offer: a personal reminder or discount before they lapse for good. Segments describe how customers behaved; they do not explain why.

الثقة: عالية · العينة: 1,363 · درجة الأثر: 58

## الأرقام الرئيسية

العرض الافتراضي للوحة المعلومات: آخر 90 يومًا من البيانات (2024-10-03 إلى 2024-12-31) مقارنة بال 90 يومًا التي سبقتها (2024-07-05 إلى 2024-10-02).

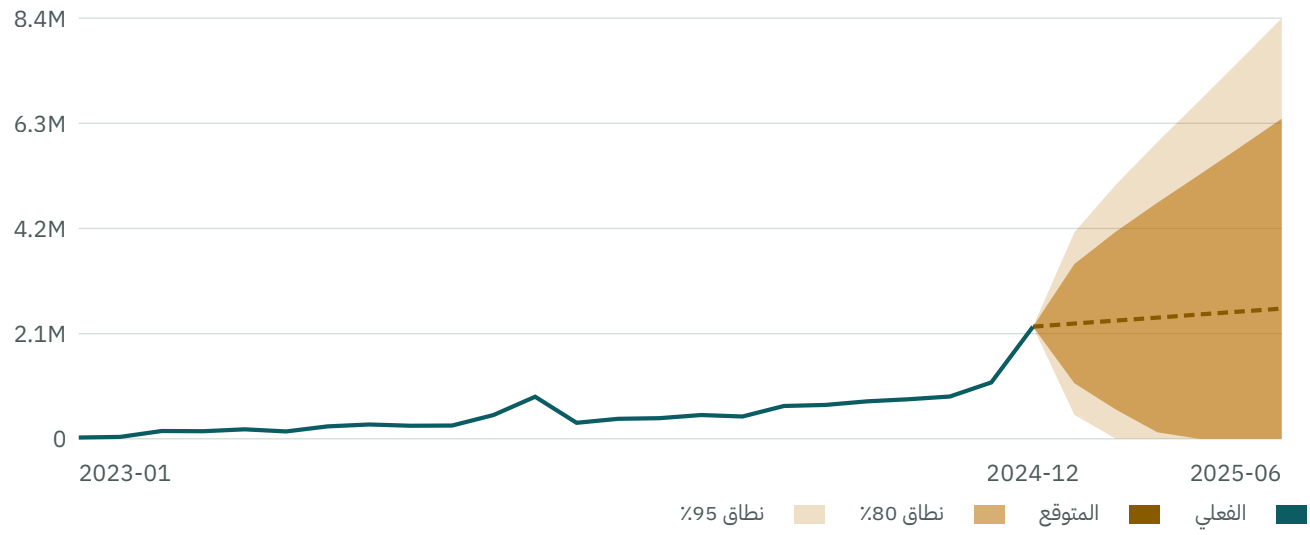
الثقة: عالية · العينة: 7,027

| المقياس | القيمة | الفترة السابقة | التغير |
|---|---|---|---|
| **صافي الإيرادات** | 4,155,778 | 2,189,144 | +89.8% |
| **الطلبات** | 7,025 | 4,111 | +70.9% |
| **متوسط قيمة الطلب** | 592 | 533 | +11.1% |
| **الوحدات** | 15,114 | 8,671 | +74.3% |
| **هامش الربح الإجمالي** | 38.1% | 38.1% | 0.0 نقطة |

## الاتجاه والتوقعات

صافي الإيرادات لكل شهر مع توقع الفترات القادمة. النطاق الأغمق هو نطاق 80٪ والأفتح هو نطاق 95٪. النموذج: AutoETS.

الثقة: متوسطة · العينة: 21,971



| الفترة | الفعلي | المتوقع | نطاق 80٪ | نطاق 95٪ |
|---|---|---|---|---|
| يناير 2024 | 319,366 | – | – | – |
| فبراير 2024 | 397,297 | – | – | – |
| مارس 2024 | 410,730 | – | – | – |
| أبريل 2024 | 473,936 | – | – | – |
| مايو 2024 | 445,315 | – | – | – |
| يونيو 2024 | 652,861 | – | – | – |
| يوليو 2024 | 675,644 | – | – | – |
| أغسطس 2024 | 747,571 | – | – | – |
| سبتمبر 2024 | 786,578 | – | – | – |
| أكتوبر 2024 | 843,975 | – | – | – |
| نوفمبر 2024 | 1,123,625 | – | – | – |
| ديسمبر 2024 | 2,235,130 | – | – | – |
| يناير 2025 | – | 2,295,213 | 1,105,838 – 3,484,588 | 476,221 – 4,114,205 |
| فبراير 2025 | – | 2,355,401 | 584,078 – 4,126,724 | 0 – 5,064,406 |
| مارس 2025 | – | 2,415,589 | 130,414 – 4,700,764 | 0 – 5,910,463 |
| أبريل 2025 | – | 2,475,777 | 0 – 5,255,981 | 0 – 6,727,731 |
| مايو 2025 | – | 2,535,965 | 0 – 5,811,919 | 0 – 7,546,104 |
| يونيو 2025 | – | 2,596,154 | 0 – 6,379,362 | 0 – 8,382,071 |

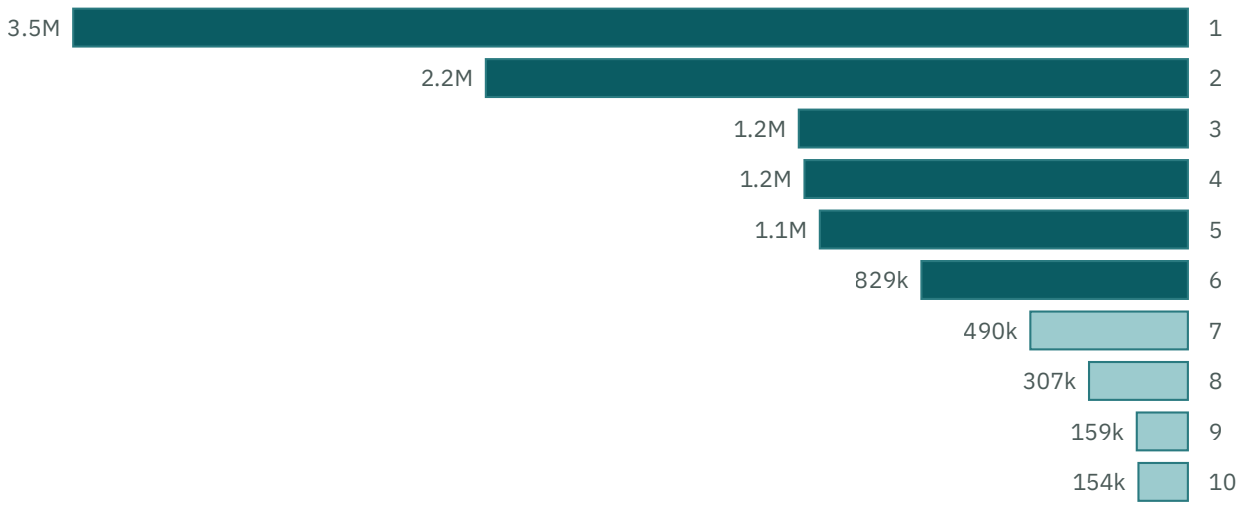
## أكبر المساهمين

المبيعات الموجبة فقط، على كامل الملف. عناصر الفئة A تشكل أول 80٪ من المبيعات، وB الـ 15٪ التالية، وC الباقي.

### المنتجات

الثقة: عالية · العينة: 21,967

A: 6 · B: 8 · C: 16 من 30 عنصرًا

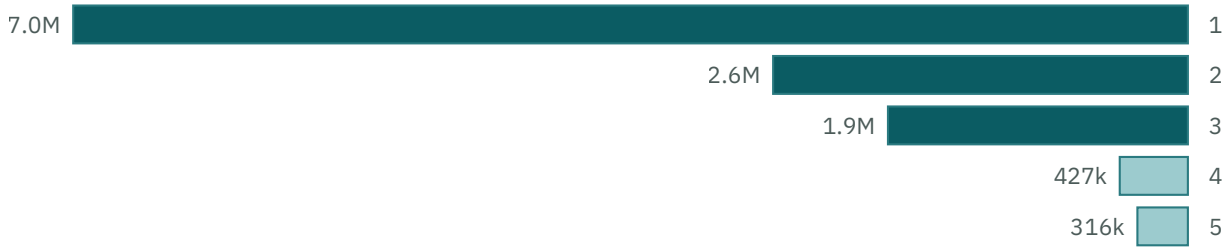


| # | العنصر | المبيعات | الحصة | التراكمي | الفئة |
|---|---|---|---|---|---|
| 1 | Item-07 | 3,470,127 | 28.5% | 28.5% | A |
| 2 | Item-12 | 2,185,541 | 17.9% | 46.4% | A |
| 3 | Item-06 | 1,211,597 | 9.9% | 56.4% | A |
| 4 | Item-19 | 1,193,537 | 9.8% | 66.2% | A |
| 5 | Item-04 | 1,145,827 | 9.4% | 75.6% | A |
| 6 | Item-02 | 829,316 | 6.8% | 82.4% | A |
| 7 | Item-16 | 490,306 | 4.0% | 86.4% | B |
| 8 | Item-27 | 307,403 | 2.5% | 88.9% | B |
| 9 | Item-13 | 159,285 | 1.3% | 90.2% | B |
| 10 | Item-17 | 153,745 | 1.3% | 91.5% | B |

## الفئات

الثقة: عالية · العينة: 21,967

A: 3 · B: 1 · C: 1 من 5 عنصرًا

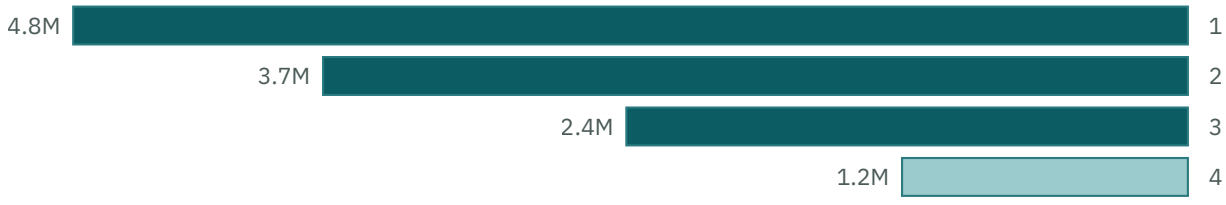



| # | العنصر | المبيعات | الحصة | التراكمي | الفئة |
|---|---|---|---|---|---|
| 1 | Furniture | 6,970,990 | 57.2% | 57.2% | A |
| 2 | Grocery | 2,593,395 | 21.3% | 78.5% | A |
| 3 | Electronics | 1,876,158 | 15.4% | 93.9% | A |
| 4 | Office | 426,768 | 3.5% | 97.4% | B |
| 5 | Apparel | 315,991 | 2.6% | 100.0% | C |

## المناطق

الثقة: عالية · العينة: 21,967

A: 3 · B: 1 · C: 0 من 4 عنصرًا




| # | العنصر | المبيعات | الحصة | التراكمي | الفئة |
|---|---|---|---|---|---|
| 1 | North | 4,803,671 | 39.4% | 39.4% | A |
| 2 | South | 3,727,294 | 30.6% | 70.0% | A |

| # | العنصر | المبيعات | الحصة | التراكمي | الفئة |
|---|---|---|---|---|---|
| 3 | East | 2,420,198 | 19.9% | 89.9% | A |
| 4 | West | 1,232,139 | 10.1% | 100.0% | B |

### منتجات بلا مبيعات حديثة

2 من أصل 30 منتجًا لم تُسجَّل لها مبيعات خلال 90 يومًا حتى 2024-12-31.

Item-30, Item-29

## أيام غير اعتيادية

فترات ابتعد فيها صافي الإيرادات كثيرًا عن مستواه المعتاد. وُجدت 8، وتُعرض أكبر 8 منها مرتبة حسب التاريخ.

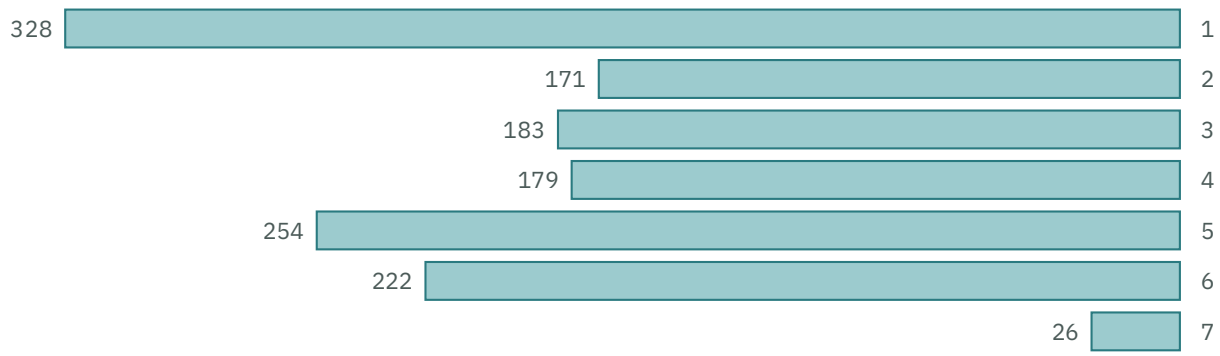
الثقة: عالية · العينة: 726

| البداية | النهاية | النوع | الفعلي | المستوى المعتاد | أكبر التغيرات في |
|---|---|---|---|---|---|
| 2023-01-09 | 2023-01-09 | ارتفاع حاد | 2,696 | 376 | South, Item-07, Furniture |
| 2023-01-27 | 2023-01-27 | انخفاض حاد | 0 | 36.77 | South, North, Item-04 |
| 2023-01-29 | 2023-01-29 | ارتفاع حاد | 3,429 | 32.26 | East, North, Item-07 |
| 2023-02-17 | 2023-02-17 | انخفاض حاد | 0 | 771 | South, North, Item-07 |
| 2023-03-14 | 2023-03-14 | ارتفاع حاد | 45,812 | 2,164 | North, South, Item-07 |
| 2023-05-09 | 2023-05-09 | انخفاض حاد | 37.50 | 7,018 | North, South, Item-07 |
| 2023-10-17 | 2023-10-17 | انخفاض حاد | 9.98 | 6,457 | North, South, Item-07 |
| 2024-02-20 | 2024-02-20 | ارتفاع حاد | 126,180 | 9,512 | North, South, Item-07 |

## شرائح العملاء

1,363 عميلًا مصنفين حسب حداثة الشراء وتكراره وقيمته. الإجراءات اقتراحات وليست نتائج مُختبرة.

الثقة: عالية · العينة: 1,363




| # | الشريحة | العملاء | الحصة من المبيعات | الإجراء المقترح |
|---|---|---|---|---|
| 1 | Champions | 328 | 50.8% | Suggestion: Reward them: early access, loyalty perks, and ask for reviews and referrals. |
| 2 | Loyal | 171 | 17.0% | Suggestion: Upsell higher-value products and invite them to a loyalty programme. |

| # | الشريحة | العملاء | الحصة من المبيعات | الإجراء المقترح |
|---|---|---|---|---|
| 3 | At risk | 183 | 13.4% | Suggestion: Win-back offer: a personal reminder or discount before they lapse for good. |
| 4 | Potential loyalists | 179 | 9.5% | Suggestion: Give a second-purchase incentive or membership to build the habit. |
| 5 | Hibernating | 254 | 5.3% | Suggestion: Low-cost reactivation with seasonal offers; do not over-invest. |
| 6 | Lost | 222 | 3.6% | Suggestion: Try one win-back campaign; otherwise stop spending marketing money on them. |
| 7 | New | 26 | 0.3% | Suggestion: Onboard well: a welcome series and a reason to buy a second time soon. |

## خطر فقدان العملاء

قُيِّم 1,170 عميلًا حسب احتمال توقفهم عن الشراء خلال 32 يومًا. تُعرض المجاميع فقط ولا يُذكر أي عميل بالاسم.

الثقة: عالية · العينة: 1,170

| المجموعة | العملاء | الإنفاق السنوي | الخسارة المتوقعة |
|---|---|---|---|
| **خطر مرتفع** | 355 | 1,473,439 | 1,279,529 |
| **خطر متوسط** | 111 | – | – |
| **خطر منخفض** | 704 | – | – |
| **أعلى 10٪ خطرًا** | 117 | 543,452 | 540,734 |

## الافتراضات والتحفظات وجودة البيانات

الإيرادات صافية بعد المرتجعات والمبالغ المستردة. الفترات بلا مبيعات تُحسب صفرًا. نطاقات التوقع هي نطاقات النموذج المختار بعد اختباره على بيانات سابقة. الارتباط لا يعني السببية.

### مشكلات جودة البيانات

لا يوجد.

### ملاحظات من التحليل

- The first or last week is only partly inside the range, so it looks smaller.

### وحدات التحليل

| الوحدة | الحالة | السبب |
|---|---|---|
| الأرقام الرئيسية | مكتملة | |
| خطر فقدان العملاء | مكتملة | |
| أكبر المساهمين | مكتملة | |
| محركات الإيرادات | مكتملة | |
| التوقع | مكتملة | |
| أيام غير اعتيادية | مكتملة | |

| الوحدة | الحالة | السبب |
|---|---|---|
| شرائح العملاء | مكتملة | |
| الاحتفاظ بالعملاء حسب الدفعة | مكتملة | |
| الاتجاه والموسمية | مكتملة | |

# ملحق: الأدلة

الاستعلامات الدقيقة وراء كل نتيجة وعدد الصفوف التي أعادها كل منها.

**النتيجة:** Headline figures (KPI table)
**يعرض:** Headline figures for the range and the window before it · **الصفوف المُعادة:** 2

```
SELECT CASE WHEN "order_date" >= DATE '2024-10-03' THEN 'current' ELSE 'previous' END AS part, count(*) AS rows, sum("revenue") AS net_revenue, sum(CASE WHEN "revenue" > 0 THEN "revenue" END) AS gross_revenue, count(DISTINCT CASE WHEN "revenue" > 0 THEN "order_id" END) AS orders, sum("quantity") AS units, sum(CASE WHEN "cost" IS NOT NULL THEN "revenue" END) AS costed_revenue, sum(CASE WHEN "revenue" IS NOT NULL THEN "cost" END) AS cost FROM data WHERE "revenue" IS NOT NULL AND "order_date" BETWEEN DATE '2024-07-05' AND DATE '2024-12-31' GROUP BY 1 ORDER BY 1
```

**النتيجة:** Headline figures (KPI table)
**يعرض:** Net revenue and orders by week · **الصفوف المُعادة:** 14

```
SELECT CAST(date_trunc('week', "order_date") AS DATE) AS period, count(*) AS rows, sum("revenue") AS net_revenue, sum(CASE WHEN "revenue" > 0 THEN "revenue" END) AS gross_revenue, count(DISTINCT CASE WHEN "revenue" > 0 THEN "order_id" END) AS orders, sum("quantity") AS units, sum(CASE WHEN "cost" IS NOT NULL THEN "revenue" END) AS costed_revenue, sum(CASE WHEN "revenue" IS NOT NULL THEN "cost" END) AS cost FROM data WHERE "revenue" IS NOT NULL AND "order_date" BETWEEN DATE '2024-10-03' AND DATE '2024-12-31' GROUP BY 1 ORDER BY 1
```

**النتيجة:** Year to date is 196.7% ahead of last year
**يعرض:** Year-to-date and prior-year-to-date net revenue · **الصفوف المُعادة:** 1

```
SELECT sum(CASE WHEN "order_date" >= DATE '2024-01-01' AND "order_date" <= DATE '2024-12-31' THEN "revenue" END) AS ytd, sum(CASE WHEN "order_date" >= DATE '2023-01-01' AND "order_date" <= DATE '2023-12-31' THEN "revenue" END) AS prior_ytd, count(*) AS rows FROM data WHERE "revenue" IS NOT NULL AND "order_date" IS NOT NULL
```

**النتيجة:** Lasting downward shift in orders from 2024-03-11
**يعرض:** Weekly revenue and orders · **الصفوف المُعادة:** 103

```
SELECT CAST(date_trunc('week', "order_date") AS DATE) AS week, sum("revenue") AS revenue, count(DISTINCT "order_id") FILTER (WHERE "revenue" > 0) AS orders FROM data WHERE "order_date" IS NOT NULL AND "revenue" IS NOT NULL AND "order_date" >= DATE '2023-01-09' AND "order_date" < DATE '2024-12-30' GROUP BY 1 ORDER BY 1
```

**النتيجة:** Net revenue for Furniture is estimated at about 1,339,349.34 next month (80% range: 588,591.74 to 2,090,106.93)
**يعرض:** Net revenue per month for Furniture · **الصفوف المُعادة:** 24

```
SELECT CAST(date_trunc('month', "order_date") AS DATE) AS period, "category" AS grp, sum("revenue") AS value, count(*) AS rows FROM data WHERE "revenue" IS NOT NULL AND "order_date" IS NOT NULL AND "category" IS NOT NULL AND "category" IN ('Furniture') GROUP BY 1, 2 ORDER BY 2, 1
```

**النتيجة:** Net revenue is estimated at about 2,295,213.14 next month (80% range: 1,105,837.97 to 3,484,588.30)
**يعرض:** Net revenue per month · **الصفوف المُعادة:** 24

```
SELECT CAST(date_trunc('month', "order_date") AS DATE) AS period, sum("revenue") AS value, count(*) AS rows FROM data WHERE "revenue" IS NOT NULL AND "order_date" IS NOT NULL GROUP BY 1 ORDER BY 1
```

**النتيجة:** Champions customers bring 50.8% of revenue
**يعرض:** Customers, revenue and averages per RFM segment · **الصفوف المُعادة:** 7

```
WITH purchases AS (SELECT "customer_id" AS customer, "order_date" AS day, "order_id" AS order_key, "revenue" AS revenue FROM data WHERE "customer_id" IS NOT NULL AND "order_date" IS NOT NULL AND "revenue" > 0), customers AS (SELECT customer, date_diff('day', max(day), DATE '2024-12-31') AS recency_days, count(DISTINCT order_key) AS frequency, sum(revenue) AS monetary, date_diff('day', min(day), DATE '2024-12-31') AS tenure_days FROM purchases GROUP BY customer), scored AS (SELECT *, CAST(1 + floor(5 * (rank() OVER (ORDER BY recency_days DESC) - 1) / count(*) OVER ()) AS INTEGER) AS r_score, CAST(1 + floor(5 * (rank() OVER (ORDER BY frequency ASC) - 1) / count(*) OVER ()) AS INTEGER) AS f_score, CAST(1 + floor(5 * (rank() OVER (ORDER BY monetary ASC) - 1) / count(*) OVER ()) AS INTEGER) AS m_score FROM customers), segmented AS (SELECT *, CASE WHEN r_score = 5 AND f_score = 5 THEN 'Champions' WHEN r_score = 5 AND f_score = 4 THEN 'Champions' WHEN r_score = 5 AND f_score = 3 THEN 'Potential loyalists' WHEN r_score = 5 AND f_score = 2 THEN 'Potential loyalists' WHEN r_score = 5 AND f_score = 1 THEN 'New' WHEN r_score = 4 AND f_score = 5 THEN 'Champions' WHEN r_score = 4 AND f_score = 4 THEN 'Champions' WHEN r_score = 4 AND f_score = 3 THEN 'Potential loyalists' WHEN r_score = 4 AND f_score = 2 THEN 'Potential loyalists' WHEN r_score = 4 AND f_score = 1 THEN 'New' WHEN r_score = 3 AND f_score = 5 THEN 'Loyal' WHEN r_score = 3 AND f_score = 4 THEN 'Loyal' WHEN r_score = 3 AND f_score = 3 THEN 'Loyal' WHEN r_score = 3 AND f_score = 2 THEN 'Hibernating' WHEN r_score = 3 AND f_score = 1 THEN 'Hibernating' WHEN r_score = 2 AND f_score = 5 THEN 'At risk' WHEN r_score = 2 AND f_score = 4 THEN 'At risk' WHEN r_score = 2 AND f_score = 3 THEN 'At risk' WHEN r_score = 2 AND f_score = 2 THEN 'Hibernating' WHEN r_score = 2 AND f_score = 1 THEN 'Hibernating' WHEN r_score = 1 AND f_score = 5 THEN 'At risk' WHEN r_score = 1 AND f_score = 4 THEN 'At risk' WHEN r_score = 1 AND f_score = 3 THEN 'At risk' WHEN r_score = 1 AND f_score = 2 THEN 'Lost' WHEN r_score = 1 AND f_score = 1 THEN 'Lost' END AS segment FROM scored) SELECT segment, count(*) AS customers, round(100.0 * sum(monetary) / sum(sum(monetary)) OVER (), 2) AS revenue_share_pct, round(sum(monetary), 2) AS revenue, round(avg(recency_days), 1) AS avg_recency_days, round(avg(frequency), 2) AS avg_orders, round(avg(monetary), 2) AS avg_spend, round(sum(monetary * 365.0 / greatest(tenure_days, 365)), 2) AS yearly_spend FROM segmented GROUP BY segment ORDER BY revenue DESC, segment
```

**النتيجة:** Revenue is up 166.5% vs the same month last year (2024-12)
**يعرض:** Net revenue by month · **الصفوف المُعادة:** 24

```
SELECT CAST(date_trunc('month', "order_date") AS DATE) AS period, sum("revenue") AS revenue, count(*) AS rows FROM data WHERE "revenue" IS NOT NULL AND "order_date" IS NOT NULL GROUP BY 1 ORDER BY 1
```

**النتيجة:** 355 customers are at high estimated churn risk
**يعرض:** Active customers at the training cutoff and how many did not buy again within 32 days · **الصفوف المُعادة:** 1

```
WITH purchases AS (SELECT "customer_id" AS customer, "order_date" AS day, "order_id" AS order_key, CAST("revenue" AS DOUBLE) AS revenue, "category" AS category, CAST("discount_pct" AS DOUBLE) AS discount FROM data WHERE "customer_id" IS NOT NULL AND "order_date" IS NOT NULL AND "revenue" > 0 AND "order_date" <= DATE '2024-10-28'), active AS (SELECT customer FROM purchases GROUP BY customer HAVING max(day) > DATE '2023-10-29'), future AS (SELECT DISTINCT "customer_id" AS customer FROM data WHERE "customer_id" IS NOT NULL AND "revenue" > 0 AND "order_date" > DATE '2024-10-28' AND "order_date" <= DATE '2024-11-29') SELECT DATE '2024-10-28' AS cutoff, 32 AS window_days, count(*) AS active_customers, count(*) FILTER (WHERE f.customer IS NULL) AS churned, round(100.0 * count(*) FILTER (WHERE f.customer IS NULL) / count(*), 2) AS churn_rate_pct FROM active a LEFT JOIN future f ON a.customer = f.customer
```

**النتيجة:** 355 customers are at high estimated churn risk
**يعرض:** Active customers at the evaluation cutoff and how many did not buy again within 32 days · **الصفوف المُعادة:** 1

```
WITH purchases AS (SELECT "customer_id" AS customer, "order_date" AS day, "order_id" AS order_key, CAST("revenue" AS DOUBLE) AS revenue, "category" AS category, CAST("discount_pct" AS DOUBLE) AS discount FROM data WHERE "customer_id" IS NOT NULL AND "order_date" IS NOT NULL AND "revenue" > 0 AND "order_date" <= DATE '2024-11-29'), active AS (SELECT customer FROM purchases GROUP BY customer HAVING max(day) > DATE '2023-11-30'), future AS (SELECT DISTINCT "customer_id" AS customer FROM data WHERE "customer_id" IS NOT NULL AND "revenue" > 0 AND "order_date" > DATE '2024-11-29' AND "order_date" <= DATE '2024-12-31') SELECT DATE '2024-11-29' AS cutoff, 32 AS window_days, count(*) AS active_customers, count(*) FILTER (WHERE f.customer IS NULL) AS churned, round(100.0 * count(*) FILTER (WHERE f.customer IS NULL) / count(*), 2) AS churn_rate_pct FROM active a LEFT JOIN future f ON a.customer = f.customer
```

**النتيجة:** 355 customers are at high estimated churn risk
**يعرض:** Averages of churners (1) and stayers (0) at the evaluation cutoff 2024-11-29 · **الصفوف المُعادة:** 2

```
WITH base AS (WITH base AS (WITH purchases AS (SELECT "customer_id" AS customer, "order_date" AS day, "order_id" AS order_key, CAST("revenue" AS DOUBLE) AS revenue, "category" AS category, CAST("discount_pct" AS DOUBLE) AS discount FROM data WHERE "customer_id" IS NOT NULL AND "order_date" IS NOT NULL AND "revenue" > 0 AND "order_date" <= DATE '2024-11-29'), feats AS (SELECT customer, date_diff('day', max(day), DATE '2024-11-29') AS recency_days, count(DISTINCT order_key) AS frequency, sum(revenue) AS monetary, date_diff('day', min(day), DATE '2024-11-29') AS tenure_days, coalesce(sum(revenue) FILTER (WHERE day > DATE '2024-08-31'), 0) AS spend_recent, coalesce(sum(revenue) FILTER (WHERE day > DATE '2024-06-02' AND day <= DATE '2024-08-31'), 0) AS spend_before, CAST(date_diff('day', min(day), max(day)) AS DOUBLE) / nullif(count(DISTINCT day) - 1, 0) AS avg_gap_days, count(DISTINCT category) AS category_count, coalesce(sum(revenue) FILTER (WHERE discount > 0), 0) / sum(revenue) AS discount_share FROM purchases GROUP BY customer) SELECT * FROM feats WHERE recency_days < 365 ORDER BY customer), future AS (SELECT DISTINCT "customer_id" AS customer FROM data WHERE "customer_id" IS NOT NULL AND "revenue" > 0 AND "order_date" > DATE '2024-11-29' AND "order_date" <= DATE '2024-12-31') SELECT base.*, CASE WHEN f.customer IS NULL THEN 1 ELSE 0 END AS churned FROM base LEFT JOIN future f ON base.customer = f.customer ORDER BY base.customer) SELECT churned, count(*) AS customers, round(avg(recency_days), 2) AS avg_recency_days, round(avg(frequency), 2) AS avg_orders, round(avg(monetary), 2) AS avg_spend, round(avg(tenure_days), 2) AS avg_tenure_days FROM base GROUP BY churned ORDER BY churned
```

**النتيجة:** 4 customer groups found
**يعرض:** Clustering inputs per customer (averages and medians) · **الصفوف المُعادة:** 1

```
WITH purchases AS (SELECT "customer_id" AS customer, "order_date" AS day, "order_id" AS order_key, "revenue" AS revenue FROM data WHERE "customer_id" IS NOT NULL AND "order_date" IS NOT NULL AND "revenue" > 0), customers AS (SELECT customer, date_diff('day', max(day), DATE '2024-12-31') AS recency_days, count(DISTINCT order_key) AS frequency, sum(revenue) AS monetary, date_diff('day', min(day), DATE '2024-12-31') AS tenure_days FROM purchases GROUP BY customer), scored AS (SELECT *, CAST(1 + floor(5 * (rank() OVER (ORDER BY recency_days DESC) - 1) / count(*) OVER ()) AS INTEGER) AS r_score, CAST(1 + floor(5 * (rank() OVER (ORDER BY frequency ASC) - 1) / count(*) OVER ()) AS INTEGER) AS f_score, CAST(1 + floor(5 * (rank() OVER (ORDER BY monetary ASC) - 1) / count(*) OVER ()) AS INTEGER) AS m_score FROM customers), segmented AS (SELECT *, CASE WHEN r_score = 5 AND f_score = 5 THEN 'Champions' WHEN r_score = 5 AND f_score = 4 THEN 'Champions' WHEN r_score = 5 AND f_score = 3 THEN 'Potential loyalists' WHEN r_score = 5 AND f_score = 2 THEN 'Potential loyalists' WHEN r_score = 5 AND f_score = 1 THEN 'New' WHEN r_score = 4 AND f_score = 5 THEN 'Champions' WHEN r_score = 4 AND f_score = 4 THEN 'Champions' WHEN r_score = 4 AND f_score = 3 THEN 'Potential loyalists' WHEN r_score = 4 AND f_score = 2 THEN 'Potential loyalists' WHEN r_score = 4 AND f_score = 1 THEN 'New' WHEN r_score = 3 AND f_score = 5 THEN 'Loyal' WHEN r_score = 3 AND f_score = 4 THEN 'Loyal' WHEN r_score = 3 AND f_score = 3 THEN 'Loyal' WHEN r_score = 3 AND f_score = 2 THEN 'Hibernating' WHEN r_score = 3 AND f_score = 1 THEN 'Hibernating' WHEN r_score = 2 AND f_score = 5 THEN 'At risk' WHEN r_score = 2 AND f_score = 4 THEN 'At risk' WHEN r_score = 2 AND f_score = 3 THEN 'At risk' WHEN r_score = 2 AND f_score = 2 THEN 'Hibernating' WHEN r_score = 2 AND f_score = 1 THEN 'Hibernating' WHEN r_score = 1 AND f_score = 5 THEN 'At risk' WHEN r_score = 1 AND f_score = 4 THEN 'At risk' WHEN r_score = 1 AND f_score = 3 THEN 'At risk' WHEN r_score = 1 AND f_score = 2 THEN 'Lost' WHEN r_score = 1 AND f_score = 1 THEN 'Lost' END AS segment FROM scored), features AS (SELECT *, monetary / frequency AS aov FROM segmented) SELECT count(*) AS customers, round(avg(recency_days), 2) AS avg_recency_days, round(quantile_cont(recency_days, 0.5), 2) AS median_recency_days, round(avg(frequency), 2) AS avg_frequency, round(quantile_cont(frequency, 0.5), 2) AS median_frequency, round(avg(monetary), 2) AS avg_monetary, round(quantile_cont(monetary, 0.5), 2) AS median_monetary, round(avg(aov), 2) AS avg_aov, round(quantile_cont(aov, 0.5), 2) AS median_aov, round(avg(tenure_days), 2) AS avg_tenure_days, round(quantile_cont(tenure_days, 0.5), 2) AS median_tenure_days FROM features
```

**النتيجة:** Net revenue: 12,183,302.00
**يعرض:** Revenue totals: net, gross sales and returns/refunds · **الصفوف المُعادة:** 1

```
SELECT count(*) AS rows, sum("revenue") AS net_revenue, sum(CASE WHEN "revenue" > 0 THEN "revenue" END) AS gross_revenue, sum(CASE WHEN "revenue" < 0 THEN -"revenue" END) AS returns_total, count(CASE WHEN "revenue" < 0 THEN 1 END) AS returns_count, count(CASE WHEN "revenue" > 0 THEN 1 END) AS sale_rows FROM data WHERE "revenue" IS NOT NULL
```

**النتيجة:** Net revenue: 12,183,302.00
**يعرض:** Distinct orders with positive revenue · **الصفوف المُعادة:** 1

```
SELECT count(DISTINCT "order_id") AS orders FROM data WHERE "revenue" > 0 AND "order_id" IS NOT NULL
```

**النتيجة:** Net revenue: 12,183,302.00
**يعرض:** Units sold and returned · **الصفوف المُعادة:** 1

```
SELECT sum("quantity") AS net_units, sum(CASE WHEN "quantity" > 0 THEN "quantity" END) AS gross_units, sum(CASE WHEN "quantity" < 0 THEN -"quantity" END) AS returned_units FROM data WHERE "quantity" IS NOT NULL
```

**النتيجة:** Net revenue: 12,183,302.00
**يعرض:** Revenue and cost on rows that have both · **الصفوف المُعادة:** 1

```
SELECT sum("revenue") AS revenue, sum("cost") AS cost FROM data WHERE "revenue" IS NOT NULL AND "cost" IS NOT NULL
```

**النتيجة:** Peak season: December
**يعرض:** Net revenue by calendar month · **الصفوف المُعادة:** 24

```
SELECT CAST(date_trunc('month', "order_date") AS DATE) AS period, sum("revenue") AS revenue, count(*) AS rows FROM data WHERE "revenue" IS NOT NULL AND "order_date" IS NOT NULL GROUP BY 1 ORDER BY 1
```

**النتيجة:** Net revenue for Electronics is estimated at about 389,080.62 next month (80% range: 157,001.64 to 621,159.59)
**يعرض:** Net revenue per month for Electronics · **الصفوف المُعادة:** 24

```
SELECT CAST(date_trunc('month', "order_date") AS DATE) AS period, "category" AS grp, sum("revenue") AS value, count(*) AS rows FROM data WHERE "revenue" IS NOT NULL AND "order_date" IS NOT NULL AND "category" IS NOT NULL AND "category" IN ('Electronics') GROUP BY 1, 2 ORDER BY 2, 1
```

**النتيجة:** 56.8% of customers return the next month
**يعرض:** Customers per cohort and month offset · **الصفوف المُعادة:** 300

```
WITH purchases AS (SELECT "customer_id" AS customer, CAST(date_trunc('month', "order_date") AS DATE) AS month FROM data WHERE "customer_id" IS NOT NULL AND "order_date" IS NOT NULL AND "order_date" < DATE '2025-01-01' AND "revenue" > 0 GROUP BY 1, 2), firsts AS (SELECT customer, min(month) AS cohort FROM purchases GROUP BY customer) SELECT f.cohort AS cohort, date_diff('month', f.cohort, p.month) AS month_offset, count(DISTINCT p.customer) AS customers FROM purchases AS p JOIN firsts AS f ON p.customer = f.customer GROUP BY 1, 2 ORDER BY 1, 2
```

**النتيجة:** Unit price is the factor most associated with revenue per line
**يعرض:** Average revenue per line by unit price (raw association, no controls) · **الصفوف المُعادة:** 9

```
SELECT round(CAST("unit_price" AS DOUBLE), 4) AS level, count(*) AS lines, round(avg("revenue"), 2) AS avg_revenue FROM data WHERE "revenue" > 0 AND "order_date" IS NOT NULL AND CAST("unit_price" AS DOUBLE) IS NOT NULL GROUP BY 1 ORDER BY 1 LIMIT 20
```

**النتيجة:** Unit price is the factor most associated with revenue per line
**يعرض:** Average revenue per line by discount (raw association, no controls) · **الصفوف المُعادة:** 6

```
SELECT round(CAST(("discount_pct" / 100.0) AS DOUBLE), 4) AS level, count(*) AS lines, round(avg("revenue"), 2) AS avg_revenue FROM data WHERE "revenue" > 0 AND "order_date" IS NOT NULL AND CAST(("discount_pct" / 100.0) AS DOUBLE) IS NOT NULL GROUP BY 1 ORDER BY 1 LIMIT 20
```

**النتيجة:** Unit price is the factor most associated with revenue per line
**يعرض:** Average revenue per line by day of the week (raw association, no controls) · **الصفوف المُعادة:** 7

```
SELECT dayname("order_date") AS level, count(*) AS lines, round(avg("revenue"), 2) AS avg_revenue FROM data WHERE "revenue" > 0 AND "order_date" IS NOT NULL AND dayname("order_date") IS NOT NULL GROUP BY 1 ORDER BY 1 LIMIT 20
```

**النتيجة:** Top category: Furniture (57.2% of sales)
**يعرض:** Positive sales by category · **الصفوف المُعادة:** 5

```
SELECT "category" AS item, sum("revenue") AS revenue, count(*) AS rows FROM data WHERE "revenue" > 0 AND "category" IS NOT NULL GROUP BY "category" ORDER BY revenue DESC, item
```

**النتيجة:** Lasting downward shift in orders from 2024-08-19
**يعرض:** Weekly revenue and orders · **الصفوف المُعادة:** 103

```
SELECT CAST(date_trunc('week', "order_date") AS DATE) AS week, sum("revenue") AS revenue, count(DISTINCT "order_id") FILTER (WHERE "revenue" > 0) AS orders FROM data WHERE "order_date" IS NOT NULL AND "revenue" IS NOT NULL AND "order_date" >= DATE '2023-01-09' AND "order_date" < DATE '2024-12-30' GROUP BY 1 ORDER BY 1
```

**النتيجة:** Net revenue for Grocery is estimated at about 353,512.74 next month (80% range: 161,571.36 to 545,454.13)
**يعرض:** Net revenue per month for Grocery · **الصفوف المُعادة:** 24

```
SELECT CAST(date_trunc('month', "order_date") AS DATE) AS period, "category" AS grp, sum("revenue") AS value, count(*) AS rows FROM data WHERE "revenue" IS NOT NULL AND "order_date" IS NOT NULL AND "category" IS NOT NULL AND "category" IN ('Grocery') GROUP BY 1, 2 ORDER BY 2, 1
```

**النتيجة:** Strongest growth and steepest decline
**يعرض:** Net revenue by month · **الصفوف المُعادة:** 24

```
SELECT CAST(date_trunc('month', "order_date") AS DATE) AS period, sum("revenue") AS revenue, count(*) AS rows FROM data WHERE "revenue" IS NOT NULL AND "order_date" IS NOT NULL GROUP BY 1 ORDER BY 1
```

**النتيجة:** Top region: North (39.4% of sales)
**يعرض:** Positive sales by region · **الصفوف المُعادة:** 4

```
SELECT "region" AS item, sum("revenue") AS revenue, count(*) AS rows FROM data WHERE "revenue" > 0 AND "region" IS NOT NULL GROUP BY "region" ORDER BY revenue DESC, item
```

**النتيجة:** Top channel: Online (33.6% of sales)
**يعرض:** Positive sales by channel · **الصفوف المُعادة:** 3

```
SELECT "channel" AS item, sum("revenue") AS revenue, count(*) AS rows FROM data WHERE "revenue" > 0 AND "channel" IS NOT NULL GROUP BY "channel" ORDER BY revenue DESC, item
```

**النتيجة:** Top product: Item-07 (28.5% of sales)
**يعرض:** Positive sales by product · **الصفوف المُعادة:** 30

```
SELECT "product" AS item, sum("revenue") AS revenue, count(*) AS rows FROM data WHERE "revenue" > 0 AND "product" IS NOT NULL GROUP BY "product" ORDER BY revenue DESC, item
```

**النتيجة:** 8 unusual days found
**يعرض:** Revenue, orders and more per day · **الصفوف المُعادة:** 713

```
SELECT "order_date" AS period, sum("revenue") AS revenue, count(DISTINCT "order_id") FILTER (WHERE "revenue" > 0) AS orders, coalesce(sum("quantity") FILTER (WHERE "quantity" > 0), 0) AS units, coalesce(avg(("discount_pct" / 100.0)), 0) AS discount_rate FROM data WHERE "order_date" IS NOT NULL AND "revenue" IS NOT NULL GROUP BY 1 ORDER BY 1
```

**النتيجة:** Revenue spike on 2024-02-20: 13.3x the expected level
**يعرض:** Revenue per day around the anomaly · **الصفوف المُعادة:** 29

```
SELECT "order_date" AS period, sum("revenue") AS revenue, count(DISTINCT "order_id") FILTER (WHERE "revenue" > 0) AS orders, coalesce(sum("quantity") FILTER (WHERE "quantity" > 0), 0) AS units, coalesce(avg(("discount_pct" / 100.0)), 0) AS discount_rate FROM data WHERE "order_date" IS NOT NULL AND "revenue" IS NOT NULL AND "order_date" >= DATE '2024-02-06' AND "order_date" < DATE '2024-03-06' GROUP BY 1 ORDER BY 1
```

**النتيجة:** Revenue spike on 2024-02-20: 13.3x the expected level
**يعرض:** Likely contributors by product: during vs before · **الصفوف المُعادة:** 30

```
WITH event AS (SELECT "product" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2024-02-20' AND "order_date" < DATE '2024-02-21' AND "product" IS NOT NULL GROUP BY 1), base AS (SELECT "product" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2024-01-23' AND "order_date" < DATE '2024-02-20' AND "product" IS NOT NULL GROUP BY 1) SELECT coalesce(event.item, base.item) AS item, round(coalesce(event.revenue, 0) / 1, 2) AS per_day_during, round(coalesce(base.revenue, 0) / 28, 2) AS per_day_before, round(coalesce(event.revenue, 0) / 1 - coalesce(base.revenue, 0) / 28, 2) AS change_per_day FROM event FULL OUTER JOIN base ON event.item = base.item ORDER BY change_per_day DESC, item
```

**النتيجة:** Revenue spike on 2024-02-20: 13.3x the expected level
**يعرض:** Likely contributors by category: during vs before · **الصفوف المُعادة:** 5

```
WITH event AS (SELECT "category" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2024-02-20' AND "order_date" < DATE '2024-02-21' AND "category" IS NOT NULL GROUP BY 1), base AS (SELECT "category" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2024-01-23' AND "order_date" < DATE '2024-02-20' AND "category" IS NOT NULL GROUP BY 1) SELECT coalesce(event.item, base.item) AS item, round(coalesce(event.revenue, 0) / 1, 2) AS per_day_during, round(coalesce(base.revenue, 0) / 28, 2) AS per_day_before, round(coalesce(event.revenue, 0) / 1 - coalesce(base.revenue, 0) / 28, 2) AS change_per_day FROM event FULL OUTER JOIN base ON event.item = base.item ORDER BY change_per_day DESC, item
```

**النتيجة:** Revenue spike on 2024-02-20: 13.3x the expected level
**يعرض:** Likely contributors by region: during vs before · **الصفوف المُعادة:** 4

```
WITH event AS (SELECT "region" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2024-02-20' AND "order_date" < DATE '2024-02-21' AND "region" IS NOT NULL GROUP BY 1), base AS (SELECT "region" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2024-01-23' AND "order_date" < DATE '2024-02-20' AND "region" IS NOT NULL GROUP BY 1) SELECT coalesce(event.item, base.item) AS item, round(coalesce(event.revenue, 0) / 1, 2) AS per_day_during, round(coalesce(base.revenue, 0) / 28, 2) AS per_day_before, round(coalesce(event.revenue, 0) / 1 - coalesce(base.revenue, 0) / 28, 2) AS change_per_day FROM event FULL OUTER JOIN base ON event.item = base.item ORDER BY change_per_day DESC, item
```

**النتيجة:** Revenue spike on 2024-02-20: 13.3x the expected level
**يعرض:** Likely contributors by channel: during vs before · **الصفوف المُعادة:** 3

```
WITH event AS (SELECT "channel" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2024-02-20' AND "order_date" < DATE '2024-02-21' AND "channel" IS NOT NULL GROUP BY 1), base AS (SELECT "channel" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2024-01-23' AND "order_date" < DATE '2024-02-20' AND "channel" IS NOT NULL GROUP BY 1) SELECT coalesce(event.item, base.item) AS item, round(coalesce(event.revenue, 0) / 1, 2) AS per_day_during, round(coalesce(base.revenue, 0) / 28, 2) AS per_day_before, round(coalesce(event.revenue, 0) / 1 - coalesce(base.revenue, 0) / 28, 2) AS change_per_day FROM event FULL OUTER JOIN base ON event.item = base.item ORDER BY change_per_day DESC, item
```

**النتيجة:** Strongest day: Tuesday
**يعرض:** Net revenue and trading days by weekday · **الصفوف المُعادة:** 7

```
SELECT dayofweek("order_date") AS weekday, sum("revenue") AS revenue, count(DISTINCT "order_date") AS days, count(*) AS rows FROM data WHERE "revenue" IS NOT NULL AND "order_date" IS NOT NULL GROUP BY 1 ORDER BY 1
```

**النتيجة:** 2 products with no sales in the last 90 days
**يعرض:** Products with no recent sales · **الصفوف المُعادة:** 2

```
WITH quiet AS (SELECT "product" AS item, max("order_date") AS last_sale FROM data WHERE "revenue" > 0 AND "order_date" IS NOT NULL AND "product" IS NOT NULL GROUP BY "product" HAVING max("order_date") < DATE '2024-10-02') SELECT q.item AS item, q.last_sale AS last_sale, sum(x."revenue") AS lifetime_revenue, sum(CASE WHEN x."order_date" > q.last_sale - 90 THEN x."revenue" END) AS run_rate FROM data AS x JOIN quiet AS q ON x."product" = q.item WHERE x."revenue" > 0 AND x."order_date" IS NOT NULL GROUP BY q.item, q.last_sale ORDER BY lifetime_revenue DESC, item
```

**النتيجة:** Revenue spike on 2023-03-14: 21.2x the expected level
**يعرض:** Revenue per day around the anomaly · **الصفوف المُعادة:** 29

```
SELECT "order_date" AS period, sum("revenue") AS revenue, count(DISTINCT "order_id") FILTER (WHERE "revenue" > 0) AS orders, coalesce(sum("quantity") FILTER (WHERE "quantity" > 0), 0) AS units, coalesce(avg(("discount_pct" / 100.0)), 0) AS discount_rate FROM data WHERE "order_date" IS NOT NULL AND "revenue" IS NOT NULL AND "order_date" >= DATE '2023-02-28' AND "order_date" < DATE '2023-03-29' GROUP BY 1 ORDER BY 1
```

**النتيجة:** Revenue spike on 2023-03-14: 21.2x the expected level
**يعرض:** Likely contributors by product: during vs before · **الصفوف المُعادة:** 30

```
WITH event AS (SELECT "product" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-03-14' AND "order_date" < DATE '2023-03-15' AND "product" IS NOT NULL GROUP BY 1), base AS (SELECT "product" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-02-14' AND "order_date" < DATE '2023-03-14' AND "order_date" NOT IN (DATE '2023-02-17') AND "product" IS NOT NULL GROUP BY 1) SELECT coalesce(event.item, base.item) AS item, round(coalesce(event.revenue, 0) / 1, 2) AS per_day_during, round(coalesce(base.revenue, 0) / 27, 2) AS per_day_before, round(coalesce(event.revenue, 0) / 1 - coalesce(base.revenue, 0) / 27, 2) AS change_per_day FROM event FULL OUTER JOIN base ON event.item = base.item ORDER BY change_per_day DESC, item
```

**النتيجة:** Revenue spike on 2023-03-14: 21.2x the expected level
**يعرض:** Likely contributors by category: during vs before · **الصفوف المُعادة:** 5

```
WITH event AS (SELECT "category" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-03-14' AND "order_date" < DATE '2023-03-15' AND "category" IS NOT NULL GROUP BY 1), base AS (SELECT "category" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-02-14' AND "order_date" < DATE '2023-03-14' AND "order_date" NOT IN (DATE '2023-02-17') AND "category" IS NOT NULL GROUP BY 1) SELECT coalesce(event.item, base.item) AS item, round(coalesce(event.revenue, 0) / 1, 2) AS per_day_during, round(coalesce(base.revenue, 0) / 27, 2) AS per_day_before, round(coalesce(event.revenue, 0) / 1 - coalesce(base.revenue, 0) / 27, 2) AS change_per_day FROM event FULL OUTER JOIN base ON event.item = base.item ORDER BY change_per_day DESC, item
```

**النتيجة:** Revenue spike on 2023-03-14: 21.2x the expected level
**يعرض:** Likely contributors by region: during vs before · **الصفوف المُعادة:** 4

```
WITH event AS (SELECT "region" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-03-14' AND "order_date" < DATE '2023-03-15' AND "region" IS NOT NULL GROUP BY 1), base AS (SELECT "region" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-02-14' AND "order_date" < DATE '2023-03-14' AND "order_date" NOT IN (DATE '2023-02-17') AND "region" IS NOT NULL GROUP BY 1) SELECT coalesce(event.item, base.item) AS item, round(coalesce(event.revenue, 0) / 1, 2) AS per_day_during, round(coalesce(base.revenue, 0) / 27, 2) AS per_day_before, round(coalesce(event.revenue, 0) / 1 - coalesce(base.revenue, 0) / 27, 2) AS change_per_day FROM event FULL OUTER JOIN base ON event.item = base.item ORDER BY change_per_day DESC, item
```

**النتيجة:** Revenue spike on 2023-03-14: 21.2x the expected level
**يعرض:** Likely contributors by channel: during vs before · **الصفوف المُعادة:** 3

```
WITH event AS (SELECT "channel" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-03-14' AND "order_date" < DATE '2023-03-15' AND "channel" IS NOT NULL GROUP BY 1), base AS (SELECT "channel" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-02-14' AND "order_date" < DATE '2023-03-14' AND "order_date" NOT IN (DATE '2023-02-17') AND "channel" IS NOT NULL GROUP BY 1) SELECT coalesce(event.item, base.item) AS item, round(coalesce(event.revenue, 0) / 1, 2) AS per_day_during, round(coalesce(base.revenue, 0) / 27, 2) AS per_day_before, round(coalesce(event.revenue, 0) / 1 - coalesce(base.revenue, 0) / 27, 2) AS change_per_day FROM event FULL OUTER JOIN base ON event.item = base.item ORDER BY change_per_day DESC, item
```

**النتيجة:** Net revenue for Apparel is estimated at about 26,532.15 next month (80% range: 18,709.89 to 36,028.48)
**يعرض:** Net revenue per month for Apparel · **الصفوف المُعادة:** 24

```
SELECT CAST(date_trunc('month', "order_date") AS DATE) AS period, "category" AS grp, sum("revenue") AS value, count(*) AS rows FROM data WHERE "revenue" IS NOT NULL AND "order_date" IS NOT NULL AND "category" IS NOT NULL AND "category" IN ('Apparel') GROUP BY 1, 2 ORDER BY 2, 1
```

**النتيجة:** Net revenue for Office is estimated at about 34,741.62 next month (80% range: 15,672.73 to 53,810.52)
**يعرض:** Net revenue per month for Office · **الصفوف المُعادة:** 24

```
SELECT CAST(date_trunc('month', "order_date") AS DATE) AS period, "category" AS grp, sum("revenue") AS value, count(*) AS rows FROM data WHERE "revenue" IS NOT NULL AND "order_date" IS NOT NULL AND "category" IS NOT NULL AND "category" IN ('Office') GROUP BY 1, 2 ORDER BY 2, 1
```

**النتيجة:** Revenue drop on 2023-05-09: 99.5% below the expected level
**يعرض:** Revenue per day around the anomaly · **الصفوف المُعادة:** 29

```
SELECT "order_date" AS period, sum("revenue") AS revenue, count(DISTINCT "order_id") FILTER (WHERE "revenue" > 0) AS orders, coalesce(sum("quantity") FILTER (WHERE "quantity" > 0), 0) AS units, coalesce(avg(("discount_pct" / 100.0)), 0) AS discount_rate FROM data WHERE "order_date" IS NOT NULL AND "revenue" IS NOT NULL AND "order_date" >= DATE '2023-04-25' AND "order_date" < DATE '2023-05-24' GROUP BY 1 ORDER BY 1
```

**النتيجة:** Revenue drop on 2023-05-09: 99.5% below the expected level
**يعرض:** Likely contributors by product: during vs before · **الصفوف المُعادة:** 28

```
WITH event AS (SELECT "product" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-05-09' AND "order_date" < DATE '2023-05-10' AND "product" IS NOT NULL GROUP BY 1), base AS (SELECT "product" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-04-11' AND "order_date" < DATE '2023-05-09' AND "product" IS NOT NULL GROUP BY 1) SELECT coalesce(event.item, base.item) AS item, round(coalesce(event.revenue, 0) / 1, 2) AS per_day_during, round(coalesce(base.revenue, 0) / 28, 2) AS per_day_before, round(coalesce(event.revenue, 0) / 1 - coalesce(base.revenue, 0) / 28, 2) AS change_per_day FROM event FULL OUTER JOIN base ON event.item = base.item ORDER BY change_per_day ASC, item
```

**النتيجة:** Revenue drop on 2023-05-09: 99.5% below the expected level
**يعرض:** Likely contributors by category: during vs before · **الصفوف المُعادة:** 5

```
WITH event AS (SELECT "category" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-05-09' AND "order_date" < DATE '2023-05-10' AND "category" IS NOT NULL GROUP BY 1), base AS (SELECT "category" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-04-11' AND "order_date" < DATE '2023-05-09' AND "category" IS NOT NULL GROUP BY 1) SELECT coalesce(event.item, base.item) AS item, round(coalesce(event.revenue, 0) / 1, 2) AS per_day_during, round(coalesce(base.revenue, 0) / 28, 2) AS per_day_before, round(coalesce(event.revenue, 0) / 1 - coalesce(base.revenue, 0) / 28, 2) AS change_per_day FROM event FULL OUTER JOIN base ON event.item = base.item ORDER BY change_per_day ASC, item
```

**النتيجة:** Revenue drop on 2023-05-09: 99.5% below the expected level
**يعرض:** Likely contributors by region: during vs before · **الصفوف المُعادة:** 4

```
WITH event AS (SELECT "region" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-05-09' AND "order_date" < DATE '2023-05-10' AND "region" IS NOT NULL GROUP BY 1), base AS (SELECT "region" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-04-11' AND "order_date" < DATE '2023-05-09' AND "region" IS NOT NULL GROUP BY 1) SELECT coalesce(event.item, base.item) AS item, round(coalesce(event.revenue, 0) / 1, 2) AS per_day_during, round(coalesce(base.revenue, 0) / 28, 2) AS per_day_before, round(coalesce(event.revenue, 0) / 1 - coalesce(base.revenue, 0) / 28, 2) AS change_per_day FROM event FULL OUTER JOIN base ON event.item = base.item ORDER BY change_per_day ASC, item
```

**النتيجة:** Revenue drop on 2023-05-09: 99.5% below the expected level
**يعرض:** Likely contributors by channel: during vs before · **الصفوف المُعادة:** 3

```
WITH event AS (SELECT "channel" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-05-09' AND "order_date" < DATE '2023-05-10' AND "channel" IS NOT NULL GROUP BY 1), base AS (SELECT "channel" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-04-11' AND "order_date" < DATE '2023-05-09' AND "channel" IS NOT NULL GROUP BY 1) SELECT coalesce(event.item, base.item) AS item, round(coalesce(event.revenue, 0) / 1, 2) AS per_day_during, round(coalesce(base.revenue, 0) / 28, 2) AS per_day_before, round(coalesce(event.revenue, 0) / 1 - coalesce(base.revenue, 0) / 28, 2) AS change_per_day FROM event FULL OUTER JOIN base ON event.item = base.item ORDER BY change_per_day ASC, item
```

**النتيجة:** Revenue drop on 2023-10-17: 99.8% below the expected level
**يعرض:** Revenue per day around the anomaly · **الصفوف المُعادة:** 29

```
SELECT "order_date" AS period, sum("revenue") AS revenue, count(DISTINCT "order_id") FILTER (WHERE "revenue" > 0) AS orders, coalesce(sum("quantity") FILTER (WHERE "quantity" > 0), 0) AS units, coalesce(avg(("discount_pct" / 100.0)), 0) AS discount_rate FROM data WHERE "order_date" IS NOT NULL AND "revenue" IS NOT NULL AND "order_date" >= DATE '2023-10-03' AND "order_date" < DATE '2023-11-01' GROUP BY 1 ORDER BY 1
```

**النتيجة:** Revenue drop on 2023-10-17: 99.8% below the expected level
**يعرض:** Likely contributors by product: during vs before · **الصفوف المُعادة:** 30

```
WITH event AS (SELECT "product" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-10-17' AND "order_date" < DATE '2023-10-18' AND "product" IS NOT NULL GROUP BY 1), base AS (SELECT "product" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-09-19' AND "order_date" < DATE '2023-10-17' AND "product" IS NOT NULL GROUP BY 1) SELECT coalesce(event.item, base.item) AS item, round(coalesce(event.revenue, 0) / 1, 2) AS per_day_during, round(coalesce(base.revenue, 0) / 28, 2) AS per_day_before, round(coalesce(event.revenue, 0) / 1 - coalesce(base.revenue, 0) / 28, 2) AS change_per_day FROM event FULL OUTER JOIN base ON event.item = base.item ORDER BY change_per_day ASC, item
```

**النتيجة:** Revenue drop on 2023-10-17: 99.8% below the expected level
**يعرض:** Likely contributors by category: during vs before · **الصفوف المُعادة:** 5

```
WITH event AS (SELECT "category" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-10-17' AND "order_date" < DATE '2023-10-18' AND "category" IS NOT NULL GROUP BY 1), base AS (SELECT "category" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-09-19' AND "order_date" < DATE '2023-10-17' AND "category" IS NOT NULL GROUP BY 1) SELECT coalesce(event.item, base.item) AS item, round(coalesce(event.revenue, 0) / 1, 2) AS per_day_during, round(coalesce(base.revenue, 0) / 28, 2) AS per_day_before, round(coalesce(event.revenue, 0) / 1 - coalesce(base.revenue, 0) / 28, 2) AS change_per_day FROM event FULL OUTER JOIN base ON event.item = base.item ORDER BY change_per_day ASC, item
```

**النتيجة:** Revenue drop on 2023-10-17: 99.8% below the expected level
**يعرض:** Likely contributors by region: during vs before · **الصفوف المُعادة:** 4

```
WITH event AS (SELECT "region" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-10-17' AND "order_date" < DATE '2023-10-18' AND "region" IS NOT NULL GROUP BY 1), base AS (SELECT "region" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-09-19' AND "order_date" < DATE '2023-10-17' AND "region" IS NOT NULL GROUP BY 1) SELECT coalesce(event.item, base.item) AS item, round(coalesce(event.revenue, 0) / 1, 2) AS per_day_during, round(coalesce(base.revenue, 0) / 28, 2) AS per_day_before, round(coalesce(event.revenue, 0) / 1 - coalesce(base.revenue, 0) / 28, 2) AS change_per_day FROM event FULL OUTER JOIN base ON event.item = base.item ORDER BY change_per_day ASC, item
```

**النتيجة:** Revenue drop on 2023-10-17: 99.8% below the expected level
**يعرض:** Likely contributors by channel: during vs before · **الصفوف المُعادة:** 3

```
WITH event AS (SELECT "channel" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-10-17' AND "order_date" < DATE '2023-10-18' AND "channel" IS NOT NULL GROUP BY 1), base AS (SELECT "channel" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-09-19' AND "order_date" < DATE '2023-10-17' AND "channel" IS NOT NULL GROUP BY 1) SELECT coalesce(event.item, base.item) AS item, round(coalesce(event.revenue, 0) / 1, 2) AS per_day_during, round(coalesce(base.revenue, 0) / 28, 2) AS per_day_before, round(coalesce(event.revenue, 0) / 1 - coalesce(base.revenue, 0) / 28, 2) AS change_per_day FROM event FULL OUTER JOIN base ON event.item = base.item ORDER BY change_per_day ASC, item
```

**النتيجة:** Revenue spike on 2023-01-09: 7.2x the expected level
**يعرض:** Revenue per day around the anomaly · **الصفوف المُعادة:** 11

```
SELECT "order_date" AS period, sum("revenue") AS revenue, count(DISTINCT "order_id") FILTER (WHERE "revenue" > 0) AS orders, coalesce(sum("quantity") FILTER (WHERE "quantity" > 0), 0) AS units, coalesce(avg(("discount_pct" / 100.0)), 0) AS discount_rate FROM data WHERE "order_date" IS NOT NULL AND "revenue" IS NOT NULL AND "order_date" >= DATE '2023-01-06' AND "order_date" < DATE '2023-01-24' GROUP BY 1 ORDER BY 1
```

**النتيجة:** Revenue spike on 2023-01-09: 7.2x the expected level
**يعرض:** Likely contributors by product: during vs before · **الصفوف المُعادة:** 2

```
WITH event AS (SELECT "product" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-01-09' AND "order_date" < DATE '2023-01-10' AND "product" IS NOT NULL GROUP BY 1), base AS (SELECT "product" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2022-12-12' AND "order_date" < DATE '2023-01-09' AND "product" IS NOT NULL GROUP BY 1) SELECT coalesce(event.item, base.item) AS item, round(coalesce(event.revenue, 0) / 1, 2) AS per_day_during, round(coalesce(base.revenue, 0) / 28, 2) AS per_day_before, round(coalesce(event.revenue, 0) / 1 - coalesce(base.revenue, 0) / 28, 2) AS change_per_day FROM event FULL OUTER JOIN base ON event.item = base.item ORDER BY change_per_day DESC, item
```

**النتيجة:** Revenue spike on 2023-01-09: 7.2x the expected level
**يعرض:** Likely contributors by category: during vs before · **الصفوف المُعادة:** 2

```
WITH event AS (SELECT "category" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-01-09' AND "order_date" < DATE '2023-01-10' AND "category" IS NOT NULL GROUP BY 1), base AS (SELECT "category" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2022-12-12' AND "order_date" < DATE '2023-01-09' AND "category" IS NOT NULL GROUP BY 1) SELECT coalesce(event.item, base.item) AS item, round(coalesce(event.revenue, 0) / 1, 2) AS per_day_during, round(coalesce(base.revenue, 0) / 28, 2) AS per_day_before, round(coalesce(event.revenue, 0) / 1 - coalesce(base.revenue, 0) / 28, 2) AS change_per_day FROM event FULL OUTER JOIN base ON event.item = base.item ORDER BY change_per_day DESC, item
```

**النتيجة:** Revenue spike on 2023-01-09: 7.2x the expected level
**يعرض:** Likely contributors by region: during vs before · **الصفوف المُعادة:** 2

```
WITH event AS (SELECT "region" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-01-09' AND "order_date" < DATE '2023-01-10' AND "region" IS NOT NULL GROUP BY 1), base AS (SELECT "region" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2022-12-12' AND "order_date" < DATE '2023-01-09' AND "region" IS NOT NULL GROUP BY 1) SELECT coalesce(event.item, base.item) AS item, round(coalesce(event.revenue, 0) / 1, 2) AS per_day_during, round(coalesce(base.revenue, 0) / 28, 2) AS per_day_before, round(coalesce(event.revenue, 0) / 1 - coalesce(base.revenue, 0) / 28, 2) AS change_per_day FROM event FULL OUTER JOIN base ON event.item = base.item ORDER BY change_per_day DESC, item
```

**النتيجة:** Revenue spike on 2023-01-09: 7.2x the expected level
**يعرض:** Likely contributors by channel: during vs before · **الصفوف المُعادة:** 1

```
WITH event AS (SELECT "channel" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-01-09' AND "order_date" < DATE '2023-01-10' AND "channel" IS NOT NULL GROUP BY 1), base AS (SELECT "channel" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2022-12-12' AND "order_date" < DATE '2023-01-09' AND "channel" IS NOT NULL GROUP BY 1) SELECT coalesce(event.item, base.item) AS item, round(coalesce(event.revenue, 0) / 1, 2) AS per_day_during, round(coalesce(base.revenue, 0) / 28, 2) AS per_day_before, round(coalesce(event.revenue, 0) / 1 - coalesce(base.revenue, 0) / 28, 2) AS change_per_day FROM event FULL OUTER JOIN base ON event.item = base.item ORDER BY change_per_day DESC, item
```

**النتيجة:** Revenue spike on 2023-01-29: 106.3x the expected level
**يعرض:** Revenue per day around the anomaly · **الصفوف المُعادة:** 22

```
SELECT "order_date" AS period, sum("revenue") AS revenue, count(DISTINCT "order_id") FILTER (WHERE "revenue" > 0) AS orders, coalesce(sum("quantity") FILTER (WHERE "quantity" > 0), 0) AS units, coalesce(avg(("discount_pct" / 100.0)), 0) AS discount_rate FROM data WHERE "order_date" IS NOT NULL AND "revenue" IS NOT NULL AND "order_date" >= DATE '2023-01-15' AND "order_date" < DATE '2023-02-13' GROUP BY 1 ORDER BY 1
```

**النتيجة:** Revenue spike on 2023-01-29: 106.3x the expected level
**يعرض:** Likely contributors by product: during vs before · **الصفوف المُعادة:** 17

```
WITH event AS (SELECT "product" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-01-29' AND "order_date" < DATE '2023-01-30' AND "product" IS NOT NULL GROUP BY 1), base AS (SELECT "product" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-01-01' AND "order_date" < DATE '2023-01-29' AND "order_date" NOT IN (DATE '2023-01-09', DATE '2023-01-27') AND "product" IS NOT NULL GROUP BY 1) SELECT coalesce(event.item, base.item) AS item, round(coalesce(event.revenue, 0) / 1, 2) AS per_day_during, round(coalesce(base.revenue, 0) / 26, 2) AS per_day_before, round(coalesce(event.revenue, 0) / 1 - coalesce(base.revenue, 0) / 26, 2) AS change_per_day FROM event FULL OUTER JOIN base ON event.item = base.item ORDER BY change_per_day DESC, item
```

**النتيجة:** Revenue spike on 2023-01-29: 106.3x the expected level
**يعرض:** Likely contributors by category: during vs before · **الصفوف المُعادة:** 5

```
WITH event AS (SELECT "category" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-01-29' AND "order_date" < DATE '2023-01-30' AND "category" IS NOT NULL GROUP BY 1), base AS (SELECT "category" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-01-01' AND "order_date" < DATE '2023-01-29' AND "order_date" NOT IN (DATE '2023-01-09', DATE '2023-01-27') AND "category" IS NOT NULL GROUP BY 1) SELECT coalesce(event.item, base.item) AS item, round(coalesce(event.revenue, 0) / 1, 2) AS per_day_during, round(coalesce(base.revenue, 0) / 26, 2) AS per_day_before, round(coalesce(event.revenue, 0) / 1 - coalesce(base.revenue, 0) / 26, 2) AS change_per_day FROM event FULL OUTER JOIN base ON event.item = base.item ORDER BY change_per_day DESC, item
```

**النتيجة:** Revenue spike on 2023-01-29: 106.3x the expected level
**يعرض:** Likely contributors by region: during vs before · **الصفوف المُعادة:** 4

```
WITH event AS (SELECT "region" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-01-29' AND "order_date" < DATE '2023-01-30' AND "region" IS NOT NULL GROUP BY 1), base AS (SELECT "region" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-01-01' AND "order_date" < DATE '2023-01-29' AND "order_date" NOT IN (DATE '2023-01-09', DATE '2023-01-27') AND "region" IS NOT NULL GROUP BY 1) SELECT coalesce(event.item, base.item) AS item, round(coalesce(event.revenue, 0) / 1, 2) AS per_day_during, round(coalesce(base.revenue, 0) / 26, 2) AS per_day_before, round(coalesce(event.revenue, 0) / 1 - coalesce(base.revenue, 0) / 26, 2) AS change_per_day FROM event FULL OUTER JOIN base ON event.item = base.item ORDER BY change_per_day DESC, item
```

**النتيجة:** Revenue spike on 2023-01-29: 106.3x the expected level
**يعرض:** Likely contributors by channel: during vs before · **الصفوف المُعادة:** 3

```
WITH event AS (SELECT "channel" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-01-29' AND "order_date" < DATE '2023-01-30' AND "channel" IS NOT NULL GROUP BY 1), base AS (SELECT "channel" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-01-01' AND "order_date" < DATE '2023-01-29' AND "order_date" NOT IN (DATE '2023-01-09', DATE '2023-01-27') AND "channel" IS NOT NULL GROUP BY 1) SELECT coalesce(event.item, base.item) AS item, round(coalesce(event.revenue, 0) / 1, 2) AS per_day_during, round(coalesce(base.revenue, 0) / 26, 2) AS per_day_before, round(coalesce(event.revenue, 0) / 1 - coalesce(base.revenue, 0) / 26, 2) AS change_per_day FROM event FULL OUTER JOIN base ON event.item = base.item ORDER BY change_per_day DESC, item
```

**النتيجة:** 15 unusual rows found
**يعرض:** Row-level checks over every row · **الصفوف المُعادة:** 1

```
WITH typical AS (SELECT "product" AS typical_item, median("unit_price") AS typical_price FROM data WHERE "unit_price" > 0 GROUP BY 1) SELECT count(*) FILTER (WHERE "quantity" = 0) AS zero_quantity, count(*) FILTER (WHERE "quantity" < 0 AND "revenue" > 0) AS negative_quantity_positive_revenue, count(*) FILTER (WHERE ("discount_pct" / 100.0) > 0.5) AS unusual_discount, count(*) FILTER (WHERE "quantity" < 0 AND "revenue" < 0) AS returns, count(*) FILTER (WHERE data."unit_price" > 0 AND (data."unit_price" >= 3.0 * typical.typical_price OR data."unit_price" <= typical.typical_price / 3.0)) AS extreme_price, count(*) AS rows FROM data LEFT JOIN typical ON data."product" = typical.typical_item
```

**النتيجة:** Churn model check: AUC 0.88 on held-out customers
**يعرض:** Active customers at the training cutoff and how many did not buy again within 32 days · **الصفوف المُعادة:** 1

```
WITH purchases AS (SELECT "customer_id" AS customer, "order_date" AS day, "order_id" AS order_key, CAST("revenue" AS DOUBLE) AS revenue, "category" AS category, CAST("discount_pct" AS DOUBLE) AS discount FROM data WHERE "customer_id" IS NOT NULL AND "order_date" IS NOT NULL AND "revenue" > 0 AND "order_date" <= DATE '2024-10-28'), active AS (SELECT customer FROM purchases GROUP BY customer HAVING max(day) > DATE '2023-10-29'), future AS (SELECT DISTINCT "customer_id" AS customer FROM data WHERE "customer_id" IS NOT NULL AND "revenue" > 0 AND "order_date" > DATE '2024-10-28' AND "order_date" <= DATE '2024-11-29') SELECT DATE '2024-10-28' AS cutoff, 32 AS window_days, count(*) AS active_customers, count(*) FILTER (WHERE f.customer IS NULL) AS churned, round(100.0 * count(*) FILTER (WHERE f.customer IS NULL) / count(*), 2) AS churn_rate_pct FROM active a LEFT JOIN future f ON a.customer = f.customer
```

**النتيجة:** Churn model check: AUC 0.88 on held-out customers
**يعرض:** Active customers at the evaluation cutoff and how many did not buy again within 32 days · **الصفوف المُعادة:** 1

```
WITH purchases AS (SELECT "customer_id" AS customer, "order_date" AS day, "order_id" AS order_key, CAST("revenue" AS DOUBLE) AS revenue, "category" AS category, CAST("discount_pct" AS DOUBLE) AS discount FROM data WHERE "customer_id" IS NOT NULL AND "order_date" IS NOT NULL AND "revenue" > 0 AND "order_date" <= DATE '2024-11-29'), active AS (SELECT customer FROM purchases GROUP BY customer HAVING max(day) > DATE '2023-11-30'), future AS (SELECT DISTINCT "customer_id" AS customer FROM data WHERE "customer_id" IS NOT NULL AND "revenue" > 0 AND "order_date" > DATE '2024-11-29' AND "order_date" <= DATE '2024-12-31') SELECT DATE '2024-11-29' AS cutoff, 32 AS window_days, count(*) AS active_customers, count(*) FILTER (WHERE f.customer IS NULL) AS churned, round(100.0 * count(*) FILTER (WHERE f.customer IS NULL) / count(*), 2) AS churn_rate_pct FROM active a LEFT JOIN future f ON a.customer = f.customer
```

**النتيجة:** Churn model check: AUC 0.88 on held-out customers
**يعرض:** Averages of churners (1) and stayers (0) at the evaluation cutoff 2024-11-29 · **الصفوف المُعادة:** 2

```
WITH base AS (WITH base AS (WITH purchases AS (SELECT "customer_id" AS customer, "order_date" AS day, "order_id" AS order_key, CAST("revenue" AS DOUBLE) AS revenue, "category" AS category, CAST("discount_pct" AS DOUBLE) AS discount FROM data WHERE "customer_id" IS NOT NULL AND "order_date" IS NOT NULL AND "revenue" > 0 AND "order_date" <= DATE '2024-11-29'), feats AS (SELECT customer, date_diff('day', max(day), DATE '2024-11-29') AS recency_days, count(DISTINCT order_key) AS frequency, sum(revenue) AS monetary, date_diff('day', min(day), DATE '2024-11-29') AS tenure_days, coalesce(sum(revenue) FILTER (WHERE day > DATE '2024-08-31'), 0) AS spend_recent, coalesce(sum(revenue) FILTER (WHERE day > DATE '2024-06-02' AND day <= DATE '2024-08-31'), 0) AS spend_before, CAST(date_diff('day', min(day), max(day)) AS DOUBLE) / nullif(count(DISTINCT day) - 1, 0) AS avg_gap_days, count(DISTINCT category) AS category_count, coalesce(sum(revenue) FILTER (WHERE discount > 0), 0) / sum(revenue) AS discount_share FROM purchases GROUP BY customer) SELECT * FROM feats WHERE recency_days < 365 ORDER BY customer), future AS (SELECT DISTINCT "customer_id" AS customer FROM data WHERE "customer_id" IS NOT NULL AND "revenue" > 0 AND "order_date" > DATE '2024-11-29' AND "order_date" <= DATE '2024-12-31') SELECT base.*, CASE WHEN f.customer IS NULL THEN 1 ELSE 0 END AS churned FROM base LEFT JOIN future f ON base.customer = f.customer ORDER BY base.customer) SELECT churned, count(*) AS customers, round(avg(recency_days), 2) AS avg_recency_days, round(avg(frequency), 2) AS avg_orders, round(avg(monetary), 2) AS avg_spend, round(avg(tenure_days), 2) AS avg_tenure_days FROM base GROUP BY churned ORDER BY churned
```

**النتيجة:** Revenue drop on 2023-01-27: 100.0% below the expected level
**يعرض:** Revenue per day around the anomaly · **الصفوف المُعادة:** 22

```
SELECT "order_date" AS period, sum("revenue") AS revenue, count(DISTINCT "order_id") FILTER (WHERE "revenue" > 0) AS orders, coalesce(sum("quantity") FILTER (WHERE "quantity" > 0), 0) AS units, coalesce(avg(("discount_pct" / 100.0)), 0) AS discount_rate FROM data WHERE "order_date" IS NOT NULL AND "revenue" IS NOT NULL AND "order_date" >= DATE '2023-01-13' AND "order_date" < DATE '2023-02-11' GROUP BY 1 ORDER BY 1
```

**النتيجة:** Revenue drop on 2023-01-27: 100.0% below the expected level
**يعرض:** Likely contributors by product: during vs before · **الصفوف المُعادة:** 15

```
WITH event AS (SELECT "product" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-01-27' AND "order_date" < DATE '2023-01-28' AND "product" IS NOT NULL GROUP BY 1), base AS (SELECT "product" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2022-12-30' AND "order_date" < DATE '2023-01-27' AND "order_date" NOT IN (DATE '2023-01-09') AND "product" IS NOT NULL GROUP BY 1) SELECT coalesce(event.item, base.item) AS item, round(coalesce(event.revenue, 0) / 1, 2) AS per_day_during, round(coalesce(base.revenue, 0) / 27, 2) AS per_day_before, round(coalesce(event.revenue, 0) / 1 - coalesce(base.revenue, 0) / 27, 2) AS change_per_day FROM event FULL OUTER JOIN base ON event.item = base.item ORDER BY change_per_day ASC, item
```

**النتيجة:** Revenue drop on 2023-01-27: 100.0% below the expected level
**يعرض:** Likely contributors by category: during vs before · **الصفوف المُعادة:** 5

```
WITH event AS (SELECT "category" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-01-27' AND "order_date" < DATE '2023-01-28' AND "category" IS NOT NULL GROUP BY 1), base AS (SELECT "category" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2022-12-30' AND "order_date" < DATE '2023-01-27' AND "order_date" NOT IN (DATE '2023-01-09') AND "category" IS NOT NULL GROUP BY 1) SELECT coalesce(event.item, base.item) AS item, round(coalesce(event.revenue, 0) / 1, 2) AS per_day_during, round(coalesce(base.revenue, 0) / 27, 2) AS per_day_before, round(coalesce(event.revenue, 0) / 1 - coalesce(base.revenue, 0) / 27, 2) AS change_per_day FROM event FULL OUTER JOIN base ON event.item = base.item ORDER BY change_per_day ASC, item
```

**النتيجة:** Revenue drop on 2023-01-27: 100.0% below the expected level
**يعرض:** Likely contributors by region: during vs before · **الصفوف المُعادة:** 4

```
WITH event AS (SELECT "region" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-01-27' AND "order_date" < DATE '2023-01-28' AND "region" IS NOT NULL GROUP BY 1), base AS (SELECT "region" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2022-12-30' AND "order_date" < DATE '2023-01-27' AND "order_date" NOT IN (DATE '2023-01-09') AND "region" IS NOT NULL GROUP BY 1) SELECT coalesce(event.item, base.item) AS item, round(coalesce(event.revenue, 0) / 1, 2) AS per_day_during, round(coalesce(base.revenue, 0) / 27, 2) AS per_day_before, round(coalesce(event.revenue, 0) / 1 - coalesce(base.revenue, 0) / 27, 2) AS change_per_day FROM event FULL OUTER JOIN base ON event.item = base.item ORDER BY change_per_day ASC, item
```

**النتيجة:** Revenue drop on 2023-01-27: 100.0% below the expected level
**يعرض:** Likely contributors by channel: during vs before · **الصفوف المُعادة:** 3

```
WITH event AS (SELECT "channel" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-01-27' AND "order_date" < DATE '2023-01-28' AND "channel" IS NOT NULL GROUP BY 1), base AS (SELECT "channel" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2022-12-30' AND "order_date" < DATE '2023-01-27' AND "order_date" NOT IN (DATE '2023-01-09') AND "channel" IS NOT NULL GROUP BY 1) SELECT coalesce(event.item, base.item) AS item, round(coalesce(event.revenue, 0) / 1, 2) AS per_day_during, round(coalesce(base.revenue, 0) / 27, 2) AS per_day_before, round(coalesce(event.revenue, 0) / 1 - coalesce(base.revenue, 0) / 27, 2) AS change_per_day FROM event FULL OUTER JOIN base ON event.item = base.item ORDER BY change_per_day ASC, item
```

**النتيجة:** Revenue drop on 2023-02-17: 100.0% below the expected level
**يعرض:** Revenue per day around the anomaly · **الصفوف المُعادة:** 28

```
SELECT "order_date" AS period, sum("revenue") AS revenue, count(DISTINCT "order_id") FILTER (WHERE "revenue" > 0) AS orders, coalesce(sum("quantity") FILTER (WHERE "quantity" > 0), 0) AS units, coalesce(avg(("discount_pct" / 100.0)), 0) AS discount_rate FROM data WHERE "order_date" IS NOT NULL AND "revenue" IS NOT NULL AND "order_date" >= DATE '2023-02-03' AND "order_date" < DATE '2023-03-04' GROUP BY 1 ORDER BY 1
```

**النتيجة:** Revenue drop on 2023-02-17: 100.0% below the expected level
**يعرض:** Likely contributors by product: during vs before · **الصفوف المُعادة:** 23

```
WITH event AS (SELECT "product" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-02-17' AND "order_date" < DATE '2023-02-18' AND "product" IS NOT NULL GROUP BY 1), base AS (SELECT "product" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-01-20' AND "order_date" < DATE '2023-02-17' AND "order_date" NOT IN (DATE '2023-01-27', DATE '2023-01-29') AND "product" IS NOT NULL GROUP BY 1) SELECT coalesce(event.item, base.item) AS item, round(coalesce(event.revenue, 0) / 1, 2) AS per_day_during, round(coalesce(base.revenue, 0) / 26, 2) AS per_day_before, round(coalesce(event.revenue, 0) / 1 - coalesce(base.revenue, 0) / 26, 2) AS change_per_day FROM event FULL OUTER JOIN base ON event.item = base.item ORDER BY change_per_day ASC, item
```

**النتيجة:** Revenue drop on 2023-02-17: 100.0% below the expected level
**يعرض:** Likely contributors by category: during vs before · **الصفوف المُعادة:** 5

```
WITH event AS (SELECT "category" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-02-17' AND "order_date" < DATE '2023-02-18' AND "category" IS NOT NULL GROUP BY 1), base AS (SELECT "category" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-01-20' AND "order_date" < DATE '2023-02-17' AND "order_date" NOT IN (DATE '2023-01-27', DATE '2023-01-29') AND "category" IS NOT NULL GROUP BY 1) SELECT coalesce(event.item, base.item) AS item, round(coalesce(event.revenue, 0) / 1, 2) AS per_day_during, round(coalesce(base.revenue, 0) / 26, 2) AS per_day_before, round(coalesce(event.revenue, 0) / 1 - coalesce(base.revenue, 0) / 26, 2) AS change_per_day FROM event FULL OUTER JOIN base ON event.item = base.item ORDER BY change_per_day ASC, item
```

**النتيجة:** Revenue drop on 2023-02-17: 100.0% below the expected level
**يعرض:** Likely contributors by region: during vs before · **الصفوف المُعادة:** 4

```
WITH event AS (SELECT "region" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-02-17' AND "order_date" < DATE '2023-02-18' AND "region" IS NOT NULL GROUP BY 1), base AS (SELECT "region" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-01-20' AND "order_date" < DATE '2023-02-17' AND "order_date" NOT IN (DATE '2023-01-27', DATE '2023-01-29') AND "region" IS NOT NULL GROUP BY 1) SELECT coalesce(event.item, base.item) AS item, round(coalesce(event.revenue, 0) / 1, 2) AS per_day_during, round(coalesce(base.revenue, 0) / 26, 2) AS per_day_before, round(coalesce(event.revenue, 0) / 1 - coalesce(base.revenue, 0) / 26, 2) AS change_per_day FROM event FULL OUTER JOIN base ON event.item = base.item ORDER BY change_per_day ASC, item
```

**النتيجة:** Revenue drop on 2023-02-17: 100.0% below the expected level
**يعرض:** Likely contributors by channel: during vs before · **الصفوف المُعادة:** 3

```
WITH event AS (SELECT "channel" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-02-17' AND "order_date" < DATE '2023-02-18' AND "channel" IS NOT NULL GROUP BY 1), base AS (SELECT "channel" AS item, sum("revenue") AS revenue FROM data WHERE "order_date" >= DATE '2023-01-20' AND "order_date" < DATE '2023-02-17' AND "order_date" NOT IN (DATE '2023-01-27', DATE '2023-01-29') AND "channel" IS NOT NULL GROUP BY 1) SELECT coalesce(event.item, base.item) AS item, round(coalesce(event.revenue, 0) / 1, 2) AS per_day_during, round(coalesce(base.revenue, 0) / 26, 2) AS per_day_before, round(coalesce(event.revenue, 0) / 1 - coalesce(base.revenue, 0) / 26, 2) AS change_per_day FROM event FULL OUTER JOIN base ON event.item = base.item ORDER BY change_per_day ASC, item
```